

Oxygen Sink Amplitude: Old Script vs Current Pipeline

Plain-language summary for comparing the Science 2024 public code, OxygenDynamics v1, and the current raw-quantification pipeline.

Short Answer

The lower sink amplitudes in the current pipeline are expected if you compare the current **NormOxySinkAmp** directly with the old Science 2024/v1 **NormOxySinkAmp**. They have the same name, but they are not the same measurement.

In the old scripts, **NormOxySinkAmp** was a processed detection-domain amplitude. In the current script, **NormOxySinkAmp** is a raw percent-change amplitude calculated from the original TIFF recording.

What The Old Science 2024 / v1 Script Did

The Science 2024 public script and the older v1 script calculated the sink event amplitude from the processed detection trace, called **Mean_OxySink_Trace_Convo**. This trace had already gone through detrending, z-scoring, spatial convolution, and temporal smoothing.

For each sink event, the old code fitted a polynomial trend line to the processed trace and measured how far the event minimum moved away from that trend line.

Old formula, simplified:

```
old NormOxySinkAmp = distance between processed event minimum and processed trend line
```

This value is useful for detection and event strength in the processed signal, but it is not a true raw fluorescence percent change.

What The Current Script Does

The current pipeline still uses the processed detection trace to find events and define event timing. However, the reported **NormOxySinkAmp** is now calculated from the original/raw TIFF trace extracted from the final detected sink region.

Current formula:

```
current NormOxySinkAmp = (mean(raw baseline) - min(raw event window)) / mean(raw baseline)
```

This means **0.10** corresponds to an approximately **10 percent raw signal drop**. If the output is shown as percent, this is multiplied by 100.

Where The Old-Style Amplitude Went

The old detection-domain amplitude is not lost. In the current pipeline it is stored separately as **DetectionOxySinkAmp**. This column is the closest match to the old Science 2024/v1 **NormOxySinkAmp**.

How To Compare Outputs Correctly

If you compare...	Use this current output
Science 2024/v1 NormOxySinkAmp	Current DetectionOxySinkAmp
Raw percent fluorescence drop	Current NormOxySinkAmp or NormOxySinkAmpPercent

Why The Current Values Can Be Lower

- The current value is a real raw percent-change measurement, not a processed z-score/trend-distance measurement.

- Denoised detection can find more events, including weaker events that were missed before. Adding weaker events can lower the mean amplitude.
- Event timing and detected sink borders can differ when detection improves, even though the final amplitude is measured from the raw TIFF.
- Correcting pixel size changes area and burden metrics strongly, but it should not directly change raw amplitude percent.

Practical Interpretation

A shift from about -25 percent in the old analysis to about -10 percent in the current analysis does not automatically mean the current pipeline is wrong. It most likely reflects that the old number and the new number describe different things.

For publication-style comparison with the Science 2024 pipeline, report that the current pipeline separates detection-domain amplitude from raw fluorescence amplitude. Use **DetectionOxySinkAmp** for legacy comparability and **NormOxySinkAmp** for raw percent-change interpretation.

Recommended Wording

In the current pipeline, oxygen sink detection may use denoised data, but signal amplitude is quantified from the original raw TIFF. The current NormOxySinkAmp is therefore a raw baseline-normalized fluorescence drop. This differs from the Science 2024/v1 scripts, where NormOxySinkAmp represented the distance of the processed detection trace from its trend line. The legacy-style detection-domain amplitude is retained as DetectionOxySinkAmp.

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